



TASK 5

EXPLORATORY DATA ANALYSIS

Titanic Dataset — Visual & Statistical Exploration

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OBJECTIVE	Extract patterns, trends, relationships and anomalies through visual and statistical exploration.
TOOLS	Python • Pandas • Matplotlib • Seaborn • Jupyter Notebook
DELIVERABLES	Jupyter Notebook + PDF Report

A professional exploratory analysis of passenger characteristics and survival patterns.

1. Dataset Overview

The Titanic dataset contains passenger-level information including survival status, passenger class, gender, age, family-related variables, ticket details, fare and embarkation information.

EDA Check	Method	Purpose
Structure	df.info()	Columns, data types and non-null records
Statistics	df.describe()	Mean, spread, quartiles and range
Categories	value_counts()	Frequency of categories and outcomes
Missing data	df.isnull().sum()	Identify incomplete fields
Relationships	Scatterplot / Pairplot	Explore variable relationships
Correlation	Heatmap	Compare numerical correlations

Data Quality

Cabin contains substantial missing data. Age and Embarked also contain missing records. These fields should be considered carefully during further analysis.

2. Visual Exploration & Observations

Age Histogram

Observation: Most passengers are concentrated in younger and middle-age groups, with fewer observations among older passengers.

Fare Boxplot

Observation: Fare contains several high-value outliers, indicating that a small number of passengers paid substantially higher fares.

Age vs Fare Scatterplot

Observation: There is no strong direct linear relationship between passenger age and fare; most observations are clustered at lower fare values.

Pairplot

Observation: The pairplot reveals relationships among numerical variables. Pclass and Fare show a noticeable association.

Correlation Heatmap

Observation: The heatmap highlights stronger and weaker linear relationships. Pclass and Fare show a notable inverse relationship.

Anomalies

High-fare observations appear as outliers, while missing values represent important data-quality issues that should be addressed before advanced analysis.

3. Survival Analysis

Survival was compared across gender, passenger class and age to identify the strongest patterns in the dataset.

Factor	Finding	Interpretation
Gender	Female passengers show higher survival outcomes.	Gender is strongly associated with survival.
Passenger class	Higher classes generally show better survival outcomes.	Class is an important survival-related factor.
Age	Survivor and non-survivor age distributions overlap.	Age alone shows a weaker relationship.

Key Insights

- Female passengers had better survival outcomes than male passengers.
- First-class passengers generally had better survival outcomes than third-class passengers.
- Age shows weaker separation between survivors and non-survivors than gender or passenger class.
- These findings describe associations in the dataset, not proof of causation.

4. Final Findings & Conclusion

- The Titanic dataset contains both numerical and categorical variables suitable for EDA.
- Descriptive statistics and frequency counts provide the baseline understanding of the dataset.
- Age is concentrated mainly in younger and middle-age ranges, while Fare contains noticeable high-value outliers.
- Missing values, especially in Cabin, are an important data-quality issue.
- Pairplot and correlation analysis reveal useful relationships among numerical features.
- Gender and passenger class are the clearest factors associated with survival outcomes.
- EDA successfully identifies patterns, trends, relationships and anomalies before further modeling.

Conclusion

Exploratory Data Analysis provides a clear understanding of the Titanic dataset before modeling. Combining statistical summaries with distribution plots, outlier analysis, relationship plots and correlation analysis reveals important survival-related patterns and data-quality issues.

Task 5 Outcome: Gained practical skill in finding patterns, trends, relationships and anomalies using Python-based EDA.

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